Tanmay Ambadkar

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Experience

Stripe New York, NY
Machine Learning Engineer Intern *May 2025 — Present*

- Architected a novel neuro-symbolic agentic workflow that combined traditional machine learning with LLMs to optimize the existing counterfeit goods detection pipeline. This improved recall by 22 points and boosted F1 score by 18 points, while simultaneously cutting LLM costs by 70% and pipeline latency by over 60%.
- Provided agent with eyes to improve performance, led the team to use internal model-training workflows and created end-to-end model training, deployment, maintenance via Flyte, Diorama.
- Boosted performance of downstream consumers of counterfeit signals by providing 36 structured insights and textual summaries with grounded evidence, along with streaming point-in-time dataset to improve risk/fraud detection and improving the merchant web

The Pennsylvania State University University Park, PA
Research Assistant *May 2023 — Present*

- Funded by US-DoD and in collaboration with Kitware & PennState IST, developed and shipped multi-objective RL frameworks for preference-based markovian rewards and non-linear multi-objective rewards. Provided end-to-end explainable artifacts by decomposing policy actions into a directed graph.
- Designing frameworks to implement Reinforcement Learning agents for optimizing operational costs of grid-integrated district energy systems, achieving a **32% cost reduction**. Created a digital twin using dynamics modeling via neural networks to further reduce training time.
- Predicted Autism Spectrum Disorder cases in children using time-series models on Electronic Health Records (EHR) data, preprocessing over 500GB of data with PySpark.

Siemens Technology and Services Remote
Research and Digitization Automation Intern *Jan. 2022 — July 2022*

- Owned the end-to-end development of a plug-and-play anomaly detection library using AutoEncoders and Explainable AI, deploying the final model on Databricks and presenting a working demo to project stakeholders.
- Engineered end-to-end forecasting workflows that achieved a **0.98 r^2 score** on internal benchmarks and built CNN-based models for a major client (Starbucks).

Siemens Technology and Services Remote
Research and Digitization Automation Intern *May 2021 — July 2021*

- Optimized the Industrial Predictive Analytics Engine (IPAE) by integrating real-time monitoring tools, reducing pipeline execution time by **30%**.
- Developed a library to forecast building occupancy and energy requirements in conjunction with an RL model for optimizing building setpoints for maximum energy savings, for Dubai Expo 2021.

Publications

[1] **Robust Adaptive Multi-Step Predictive Shielding. T. Ambadkar**, D. Chudiwal, G. Anderson, and A. Verma. *Accepted at International Conference on Learning Representations (ICLR) 2026; Accepted Student Abstract at AAAI 2026.* [Paper] [Project Page]

[2] **AutoSpec: Automating the Refinement of Reinforcement Learning Specifications. T. Ambadkar**, Djordje Zikelic, and A. Verma. *Accepted at International Conference on Learning Representations (ICLR); Accepted at Post-AI Formal Methods at AAAI-2026; Poster at PLDI SRC*, 2024. [Paper]

[3] **Safer Policies via Affine Representations using Koopman Dynamics. Tanmay Ambadkar**, Darshan Chudiwal, Greg Anderson, and Abhinav Verma. *Accepted at NASA Formal Methods, 2026.*

[4] **Preference Conditioned Multi-Objective Reinforcement Learning. T. Ambadkar**, S. Panda, S. Kale, A. Verma, and J. Dodge. *In Submission at NeurIPS 2026.* [Paper] [Project Page]

[5] **Optimizing Expected Utility in Multi Objective Reinforcement Learning. Tanmay Ambadkar**, Raghav Kumar, Sourav Panda, Shreyash Kale, Jonathan Dodge and Abhinav Verma. *In Submission at NeurIPS 2026.*

[6] **On the Structure of Reinforcement Learning Policies: Recovery from Synthetic Observations via Geometric Coverage. Tanmay Ambadkar** and Abhinav Verma. *In Submission at NeurIPS 2026.*

[7] **EnzymeLM: Chemistry-Native Language Modeling for Balanced Enzymatic Reaction Prediction. Tanmay Ambadkar**, Mohit Anand, Abhinav Verma and Costas Maranas. *In Submission at NeurIPS 2026.*

[8] **Specification Guided Reinforcement Learning. T. Ambadkar**, *Accepted at AAAI Doctoral Consortium 2026*

[9] **Scaling Strategy, Not Compute: A Stand-Alone, Open-Source StarCraft II Benchmark for Accessible RL Research.** Sourav Panda, Shreyas Kale, **Tanmay Ambadkar**, Abhinav Verma, and Jonathan Dodge. *In Submission to NeurIPS 2026.*

[10] **MIXTAPE: Middleware for Interactive XAI with Tree-Based AI Performance Evaluation.** Brian Hu, Jonathan Dodge, Abhinav Verma, **Tanmay Ambadkar**, Sourav Panda, Sujay Koujalgi, Aashish Chaudhary, Brianna Major, and Bryon Lewis. Accepted NATO, *SISO SIMposium*, 2024, 2025. [Abstract]

[11] **Optimizing Operational Costs in Combined Heat and Power Integrated District Heating Systems.** Saranya Anbarasu, **Tanmay Ambadkar**, Rosina Adhikari, Kathryn Hinkelman, Zhanwei He, Wangda Zuo, and Ardeshir Moftakhari. *SimBuild*, 2024.[Paper]

[12] **A Simple Fast Resource-efficient Deep Learning for Automatic Image Colorization. Tanmay Ambadkar** and Jignesh S. Bhatt. *Color and Imaging Conference (CIC)*, 2023.[Paper]

[13] **Discrete Sequencing for Demand Forecasting: A novel data sampling technique for time series forecasting.** N. Menon*, S. Saboo*, **Tanmay Ambadkar***, and U. Uppili. *International Conference on Intelligent Data Science Technologies and Applications (IDSTA)*, 2022.[Paper]

[14] **Deep reinforcement learning approach to predict head movement in 360° videos. Tanmay Ambadkar** and Pramit Mazumdar. *Proc. IS&T Int'l. Symp. on Electronic Imaging*, 2022.[Paper]

Education

The Pennsylvania State University — University Park, PA
Ph.D. in Computer Science and Engineering; GPA: 3.56 — *Aug. 2022 – May 2027 (Expected)*

Indian Institute of Information Technology, Vadodara — Gujarat, India
B.Tech in Computer Science and Engineering; GPA: 9.4/10 — *Aug. 2018 – May 2022*

Teaching Experience

The Pennsylvania State University — University Park, PA
Teaching Assistant

- CMPSC 448: Machine Learning and Algorithmic AI
- CMPSC 221: Object Oriented Design & Web Programming
- CMPEN 270: Digital Design: Theory and Practice

Indian Institute of Information Technology, Vadodara — Gujarat, India
Teaching Assistant

- CS201: Introduction to Data Structures

Technical Skills

Programming Languages: Python, JavaScript, C++
Frameworks: PyTorch, TensorFlow, Stable-Baselines3, Scikit-learn, Flask, PySpark, Flyte, Diorama, LangChain, Mongo, Kafka
Topics: Reinforcement Learning, AI Safety, Formal Methods, Multi-Objective RL, Time-Series Analysis, Supervised Fine-tuning, RLVR, Agentic systems